

ACTIVE MODE MAP · SCM-PIT · MULTI75G

Reconstructing the full complex spectrum

amplitude, phase, bias dependence, and honest error bars

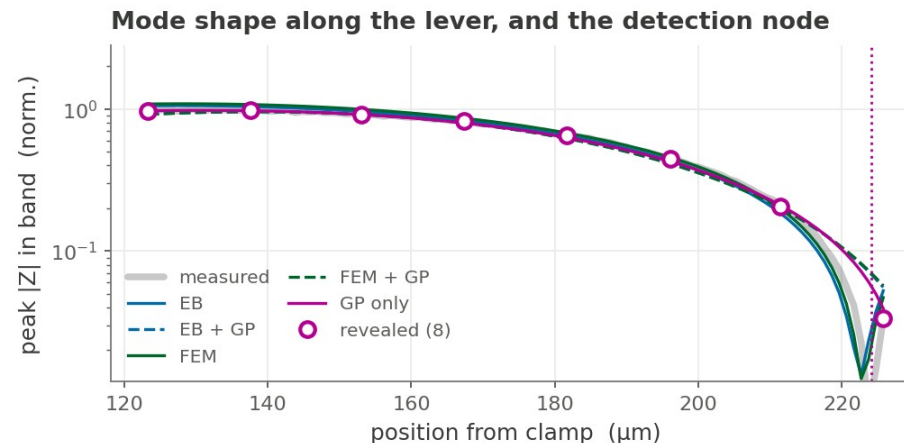
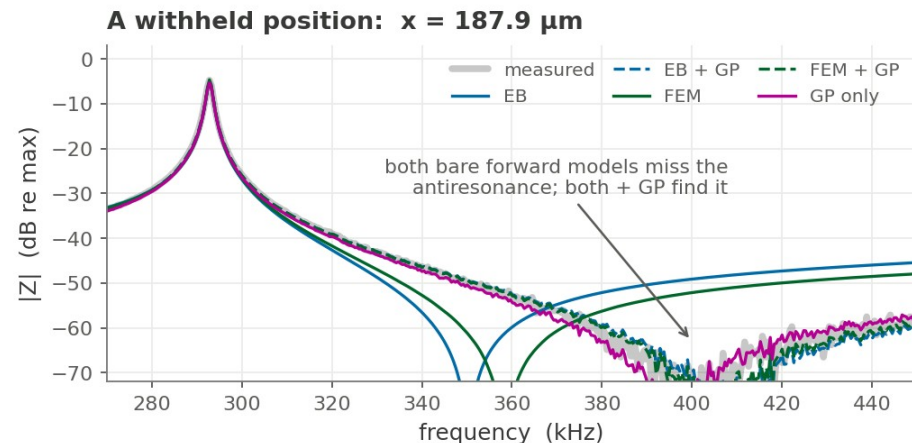
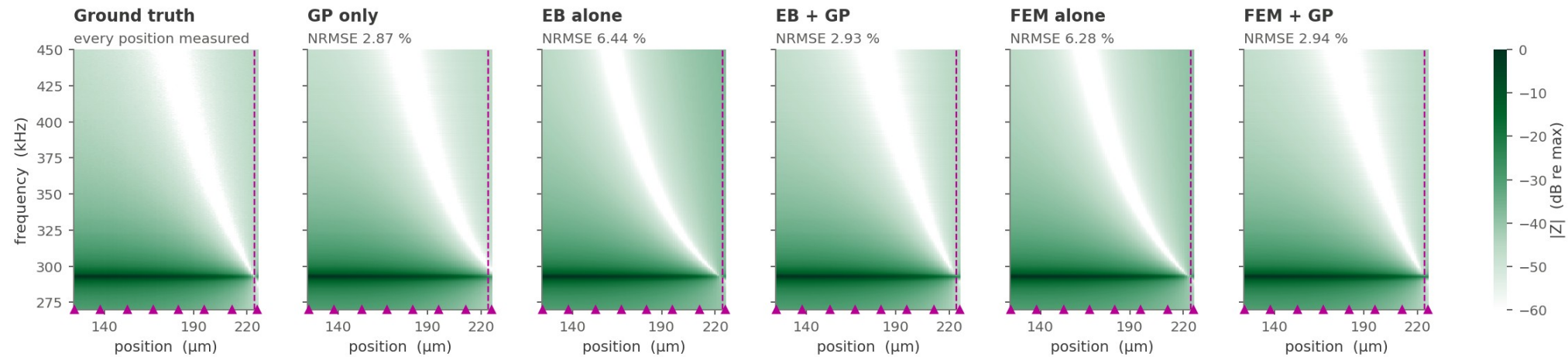
Both forward models (two-segment Euler-Bernoulli and the geometry-faithful FEM), each bare and with a discrepancy GP, and a model-light GP baseline — reconstructing the mode-A contact resonance from 8 of 101 measured positions and now carrying the phase channel, not only log-amplitude.

Dense_Grid_A · 101 positions × 412 frequencies · 14 August 2026

Amplitude: 8 of 101 positions reproduce the map to 2.9 %

Bare physics 6.3-6.4 %; with the GP either model reaches 2.9 %, the same as the GP alone. Both libraries were rebuilt here and reproduce the published log-amplitude arms exactly (EB 6.53 / 2.41, FEM 6.31 / 1.94, GP 3.41 %).

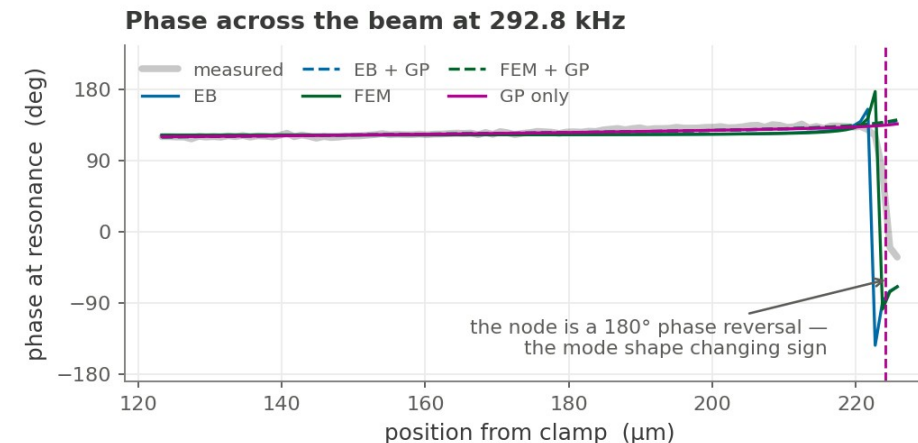
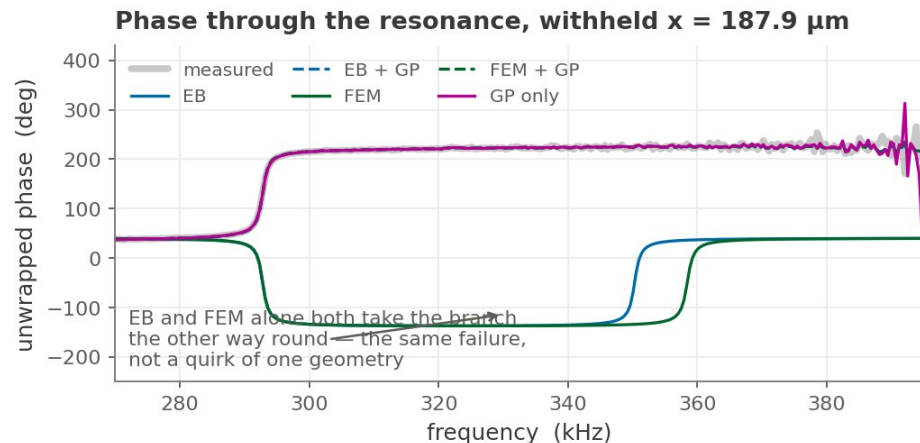
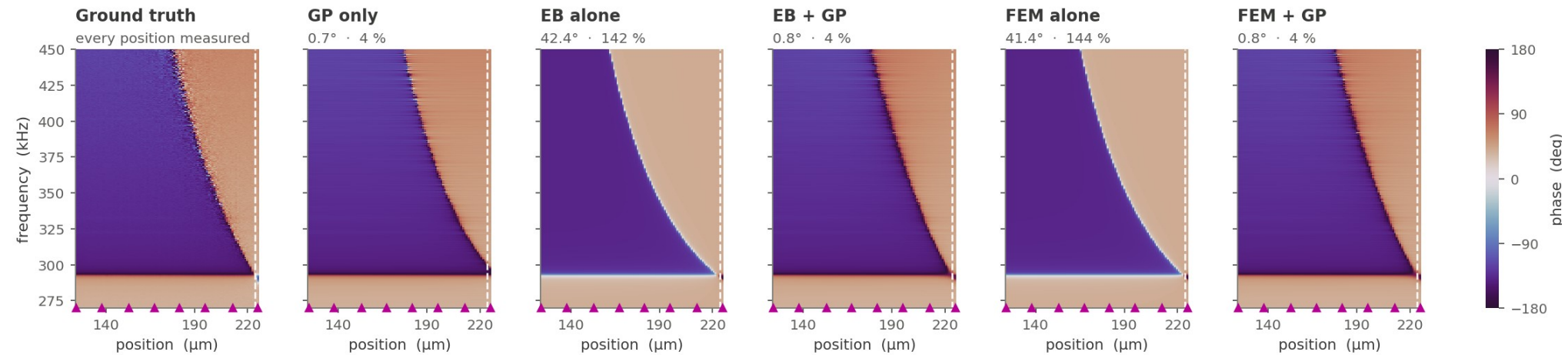
It earns its place at the node. EB + GP and FEM + GP both land 0.15 μm from the true 224.09 μm ; bare EB is 1.8 μm out, bare FEM 1.0 μm , and the GP alone never finds a crossing in span — 3 μm out at every budget from 4 to 30 positions.



Phase: both forward models get it wrong the same way

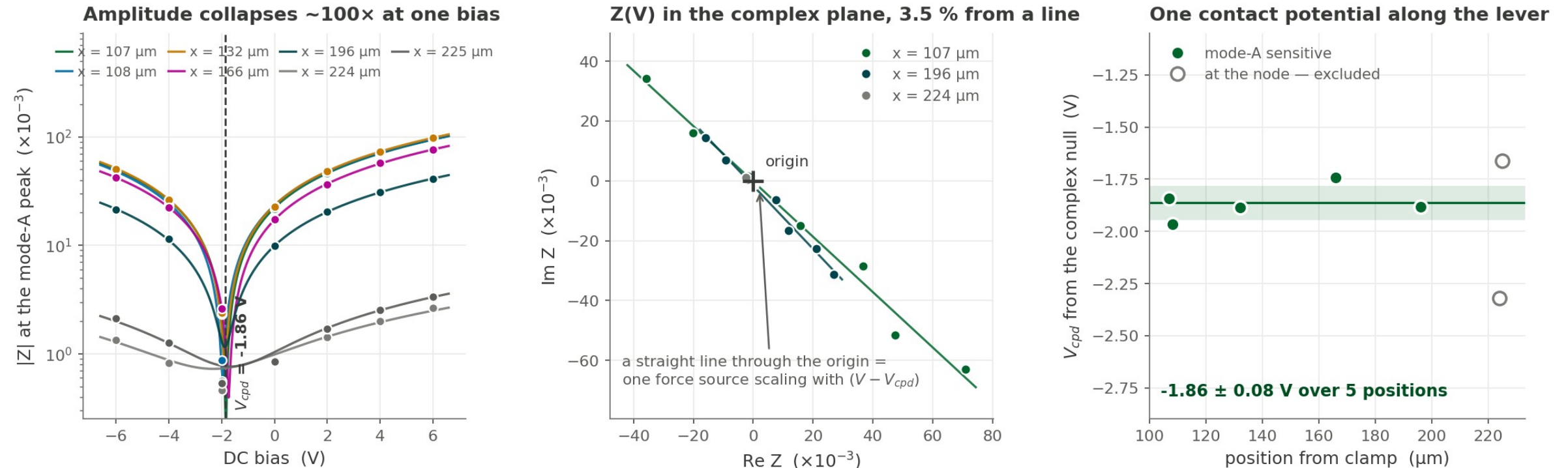
EB alone 42.4°, FEM alone 41.4° — 142 % and 144 % on the complex field, worse than predicting zero. One complex gain absorbs the detector's reference phase, so what is left is model error. Two unrelated models, contact stiffnesses 3× apart, fail by the same 41–42° and take the antiresonance branch the same wrong way.

Not a mis-set resonance either. Refitting f_{res} and damping against the complex residual only makes it worse — the discrepancy is structural.



The bias response is large, and it is electrostatic

7 biases (−6 ... +6 V) × 7 positions, 1000 nN, 1 V AC electrical drive on the tip. Each spectrum is read at its OWN resonance peak.



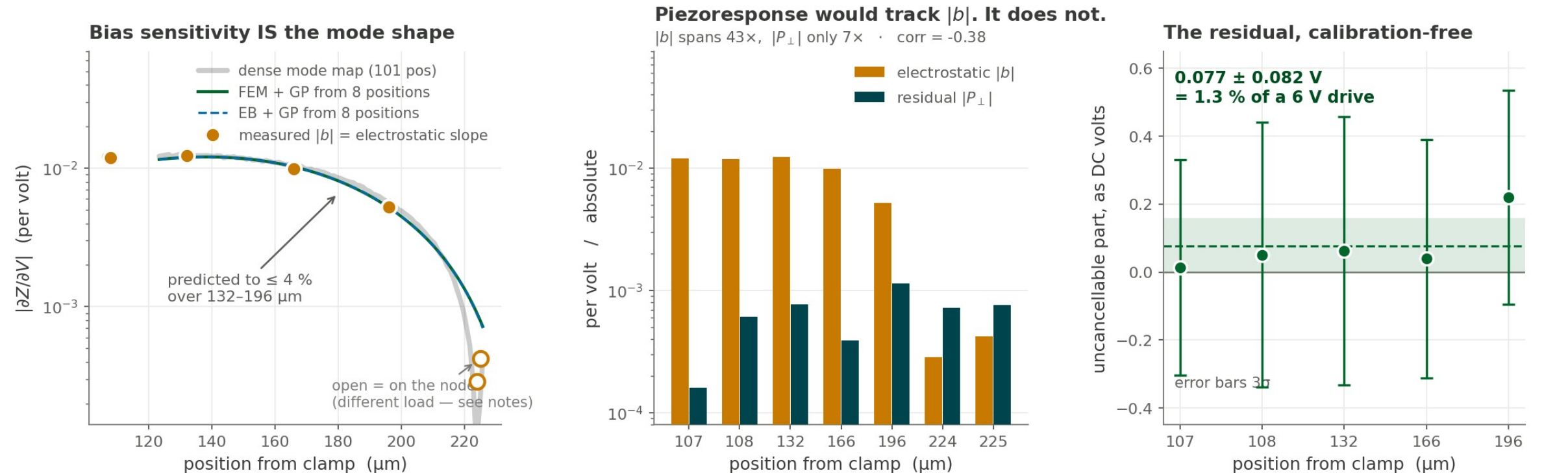
A 100× cancellation in the middle of the sweep, not at an end. Monotonic drift cannot do that, and the bias order was identical at every position — so the V-shape is the evidence that this is bias and not time.

Z(V) is a straight line through the origin to 3.5-4.4 %. That is the signature of one force source scaling with $(V - V_{cpd})$. $V_{cpd} = -1.86 \pm 0.08 \text{ V}$, flat along the lever — a tip-sample property, reproduced at five independent engages.

Q = 206 ± 4 at V = 0, and f_{res} moves less than 1 kHz across ±6 V. Still: randomise the condition order next time — this effect was big enough to be unambiguous; a 10 % effect would not be.

The mode map predicts the bias sensitivity

Splitting $Z(V)$ into the part that cancels (the electrostatic slope b) and the part that cannot ($|P_{\perp}|$, the perpendicular distance from the origin to the fitted line).



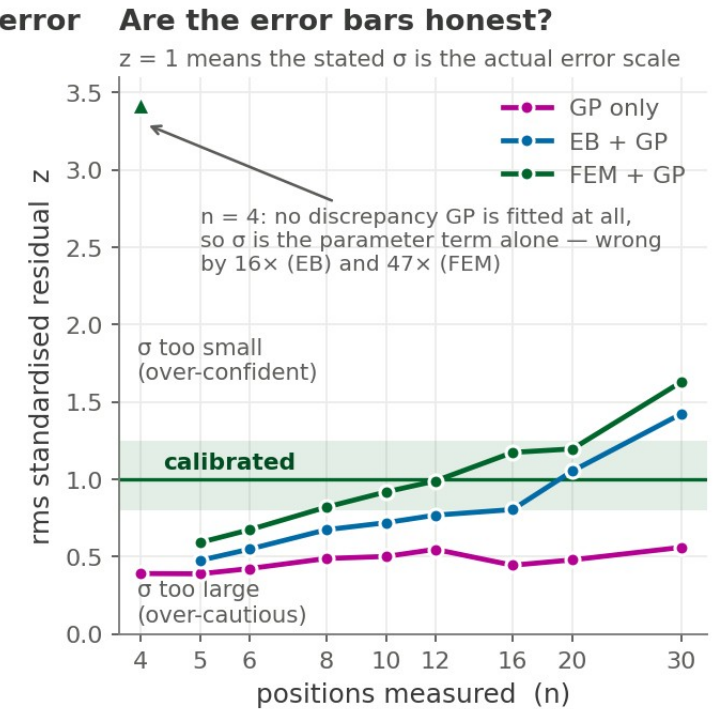
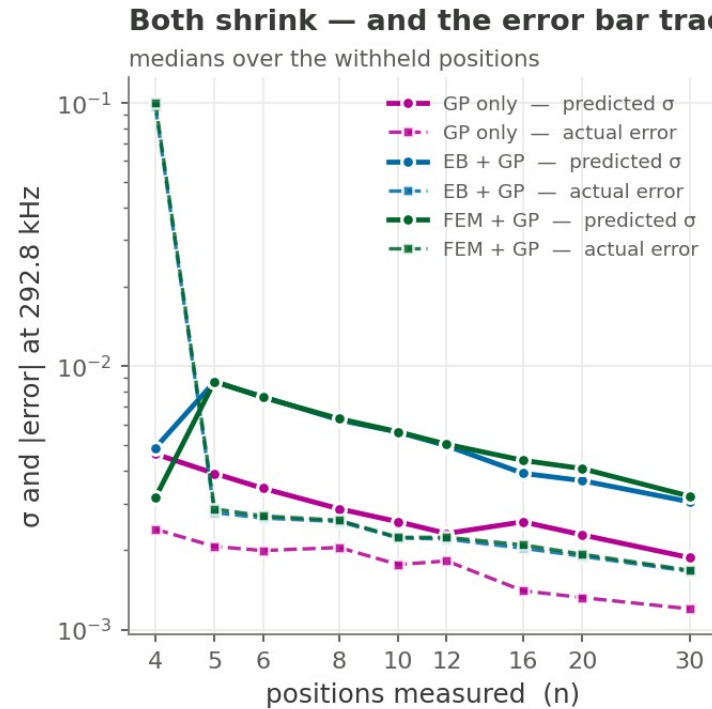
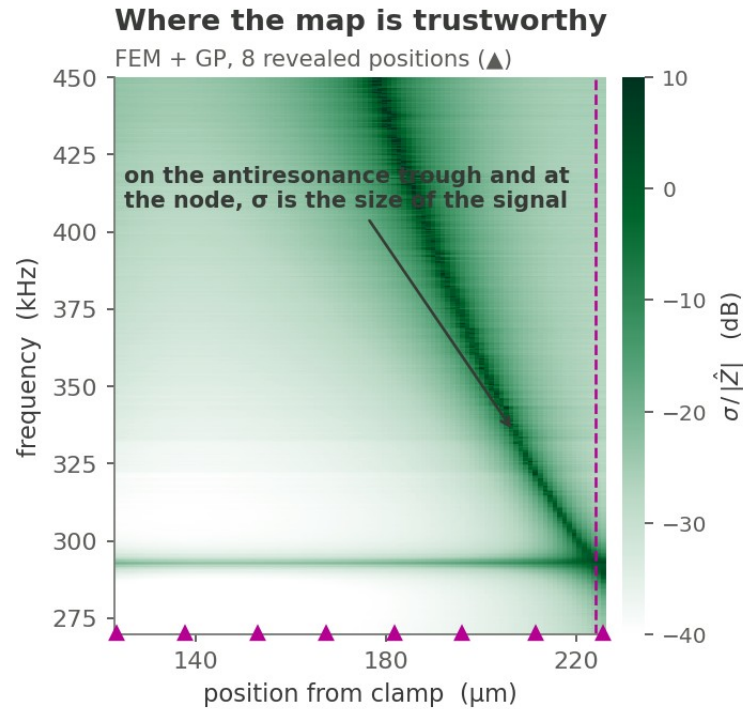
$|\partial Z/\partial V|$ along the lever IS the mode-A shape. The 8-position FEM + GP reconstruction predicts the measured electrostatic slope to $\leq 4\%$ over 132-196 μm with a single free scale — the sparse map is a calibration for the bias experiment.

The residual is instrumental, not tip-sample. $|b|$ varies 43 $\times$ along the beam; $|P_{\perp}|$ only 7 $\times$, and $\text{corr}(\log|P_{\perp}|, \log|b|) = -0.38$. A piezoresponse is driven at the tip like the electrostatic force, so it must be detected through the same mode shape — it would have collapsed at the node too.

No measurable piezoresponse: $0.077 \pm 0.082 \text{ V}$ equivalent DC offset, 1.3 % of a 6 V drive. The two positions at 224/225 μm are significant (P/SE ≈ 9 -10) only because the detection has nulled $|b|$ there.

How much of the map do we actually believe?

σ is the predictive sd of one real component of Z . Noise floor, gain, GP length scale and per-frequency variance all come from the revealed block; the withheld positions only ever score.



$\sigma \approx |Z|$ exactly on the features worth measuring. The antiresonance trough and the node are where the signal is smallest and the reconstruction least constrained. The discrepancy GP carries $\sim 4\times$ more of σ there than the parameter fit does.

Both physics arms start cautious and end over-confident. z runs $0.59 \rightarrow 0.99 \rightarrow 1.63$ for FEM + GP and $0.48 \rightarrow 0.77 \rightarrow 1.42$ for EB + GP over $n = 5 \rightarrow 12 \rightarrow 30$, while GP only sits at 0.4–0.6 throughout. At $n = 4$ no discrepancy GP is fitted and σ is wrong by $16\times / 47\times$.

What is new, what is missing, what is next

New in this pass

- Both libraries re-stitched to KEEP the complex channel — the published builds discarded the phase (FEM at `np.abs()`, EB by storing `log|z|` only). Each reproduces its published log-amplitude arm exactly.
- All six arms now reconstruct amplitude and phase, with a per-component predictive σ split into a parameter term and a discrepancy term.
- The bias series is re-analysed on its own resonance peak, and tied to the sparse mode map for the first time.

Correction to an earlier claim

- An earlier draft of this deck said the EB library could not be rebuilt because `eb_models.py` and `geometry.py` were missing. They are not missing — they live in `EB-Solver-CResonance/fem_eb_comparison/py`, not in that repo's `src/`, which holds only the `eb_cr_afm` package.
- The EB arm is fully rebuildable (96×4 rungs, 6.3 min) and is in this deck. Nothing else in the earlier numbers was affected.
- Still out of scope: mode B, the nominal-axis control, and any acquisition strategy other than equispaced.

Next

- Fit the physics on the complex field from the start. Phase is the sensitive channel for `f_res` and `Q`; amplitude is the sensitive channel for the mode shape.
- Treat the shared 41° phase error as the open physics question — two unrelated forward models failing identically points at the drive or detection model, not the beam.
- Re-derive the GP variance so σ stays calibrated past $n = 12$, and randomise the condition order in the next bias series.

Everything here is leakage-safe: each reconstructor sees only the revealed indices, their spectra and the frequency axis. Scripts: `libraries/build_femlib_cplx.py`, `libraries/buildlib_cplx.py`, `src/cplxrec.py`, `src/bias.py`, `figures/d_*.py`.